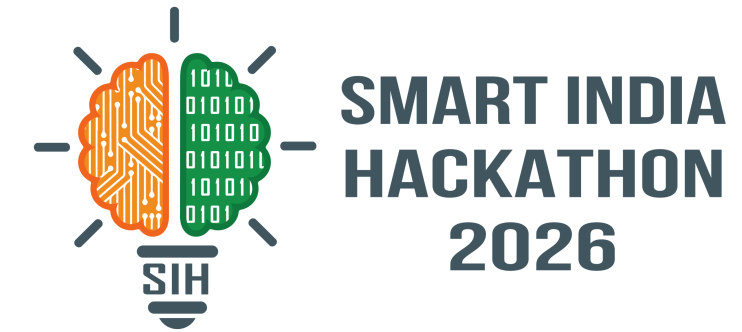
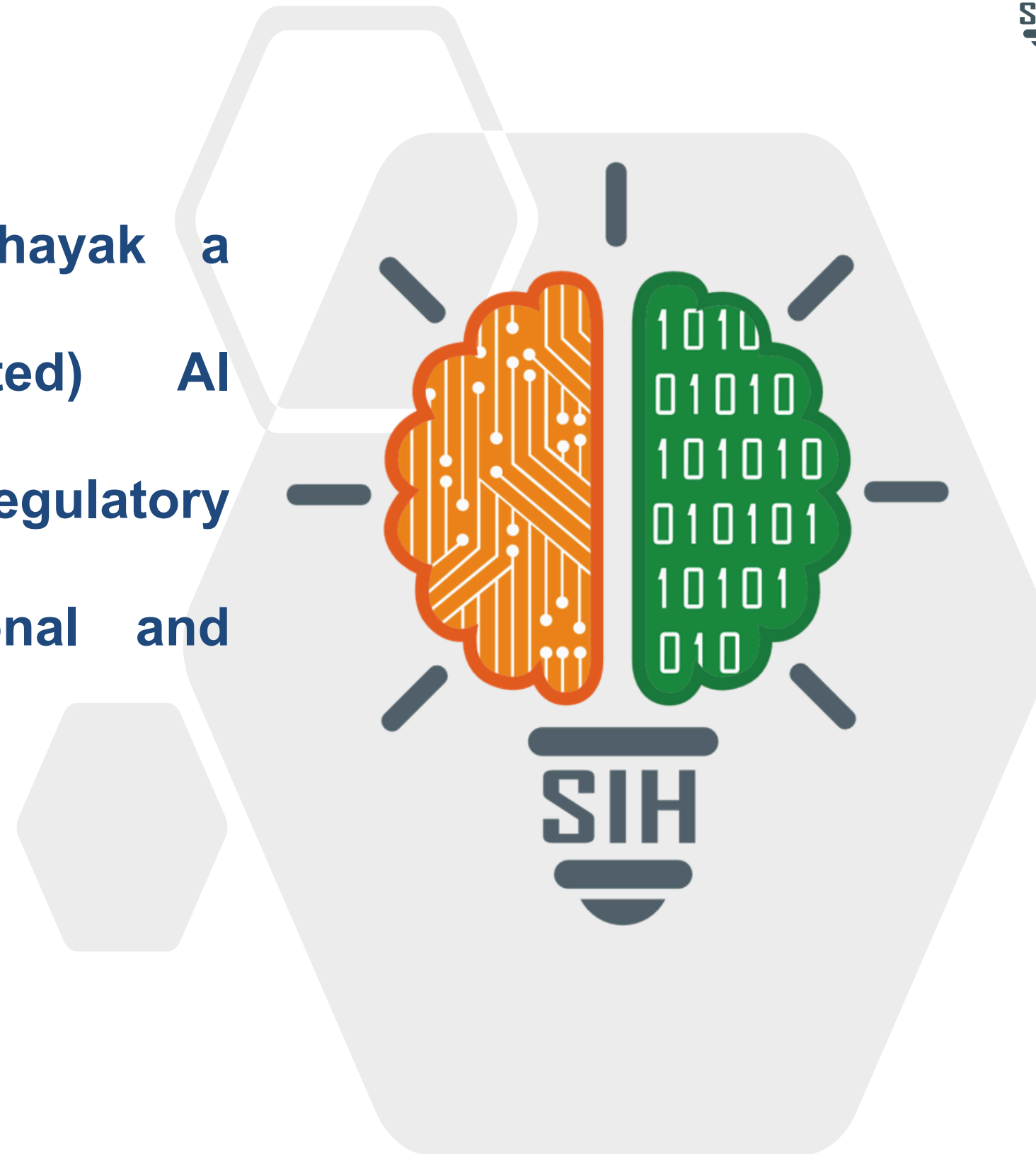


SMART INDIA HACKATHON 2026



- Problem Statement ID –**SIH26045**
- Problem Statement Title-**IP-SAKTI Sahayak** a multilingual, RAG-based (source-cited) AI assistant for Intellectual Property and regulatory guidance in Ayurveda, across national and international regimes.
- Theme-**MedTech / BioTech / HealthTech**
- PS Category- **Software**
- Team ID-
- Team Name (Registered on portal) - **Byte Us**





Detailed Explanation of the Proposed Solution

PROBLEM :

Navigating lengthy Ayurvedic Acts and Rules makes IP research complex and time-consuming, leaving non-legal users struggling to identify provisions relevant to their product..

SOLUTION:

IP-Shakti uses AI + Legal RAG to understand the product, identify the applicable IP domain and jurisdiction, retrieve the exact provisions, and explain them in plain language.

How It Addresses the Problem

One product, many laws : Matches the right IP and regulatory regimes to each product, not a one-size-fits-all answer.

Product Classification : Sorts the formulation as classical, proprietary, new drug, phytopharmaceutical, Aahar, or cosmetic - first.

From legal text to practical guidance : Turns statutory language into plain-language next steps.

Safer and more trustworthy AI : Every answer is source-cited and can abstain when the evidence is uncertain.

Innovation And Uniqueness

Legal-Aware Knowledge Structuring

Sections, clauses, legal hierarchy and metadata are preserved during ingestion, so the system retrieves legally meaningful context.

Cross-Domain Legal Retrieval

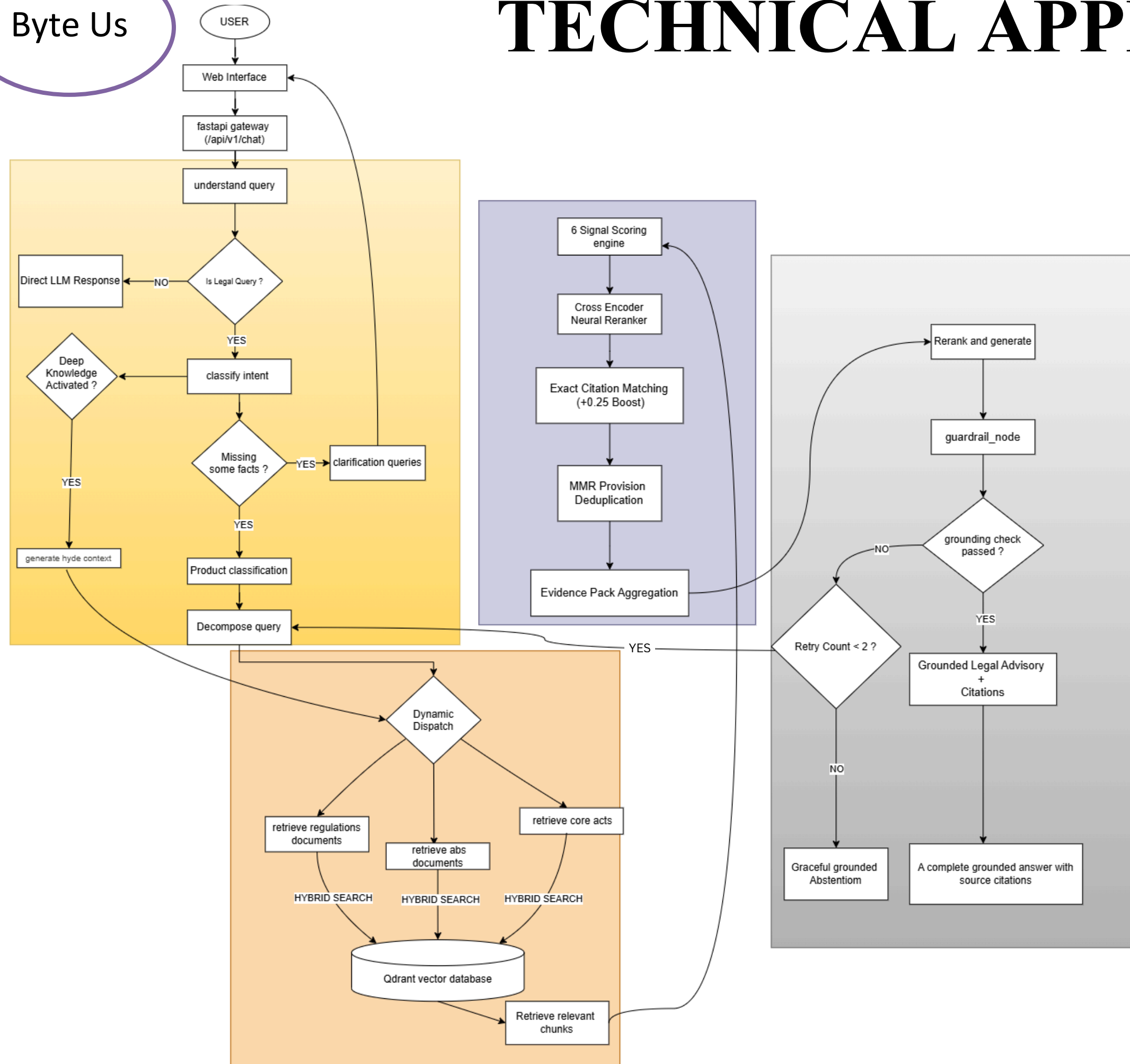
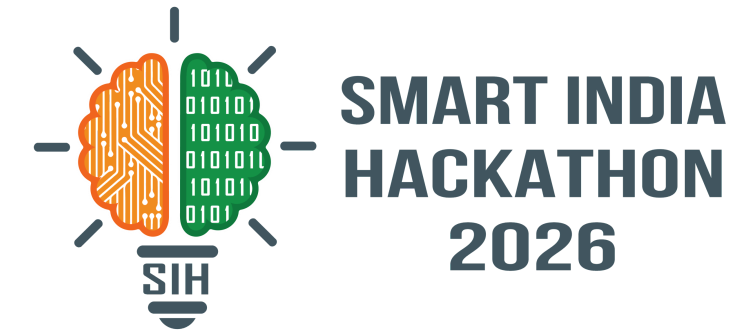
A single query can be decomposed and retrieved across Patents + GI + Biodiversity/ABS , International frameworks,

Source-Grounded Reasoning

The final answer is built from retrieved authoritative provisions, making the reasoning traceable and verifiable instead of purely generative.

Byte Us

TECHNICAL APPROACH



Complete Flowchart Link : [LINK](#)

TECHNOLOGY STACK :

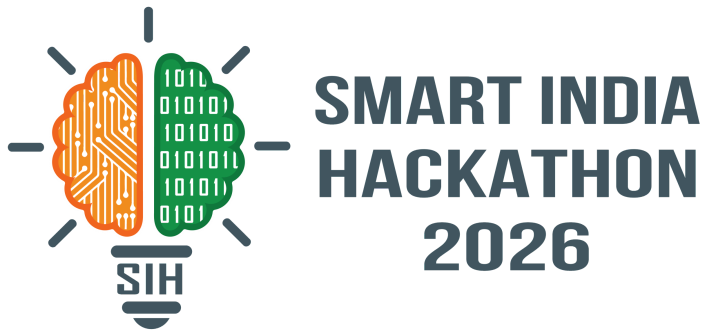
FastAPI Qdrant supabase

LangGraph LangChain

Hugging Face TANSTACK React JS



FEASIBILITY AND VIABILITY



FEASIBILITY OF IDEA

Technical Feasibility :

Built using LLMs, RAG, multi-agent orchestration, vector databases, and structured legal retrieval.

Data Feasibility : Corpus can be created from authoritative Indian IP, AYUSH, biodiversity, and regulatory sources.

Operational Feasibility: The system can automate ingestion, retrieval while uncertain cases are routed for clarification or human review.

CHALLENGES & THEIR OVERCOMINGS

- 1

Gated Data Access: Complete TKDL database remains proprietary and classified, restricted to institutional CSIR/IPO agreements only.
- ✓

Public-Domain Proxy Corpus: Ingested 37+ authoritative open sources like API monographs and classical texts, with BYO-credential vault for institutional access.
- 2

Language & Term Ambiguity: Vernacular and regional botanical names risk being dropped during query classification, causing missed prior-art checks.
- ✓

Decoupled Botanical Terms: An independent regex normalizer maps common and classical names to standard scientific binomials before retrieval starts.
- 3

Legal & Jurisdictional Risk: Generating fake legal references and mixing Indian Patent law with foreign FDA/EPO rules risk critical compliance errors.
- ✓

Air-Gapped Jurisdiction Toggles: Dedicated prompt templates and payload filters keep Indian and international statutes completely separate during retrieval, preventing cross-contamination.

National Sovereignty

Strengthens India's defense of traditional knowledge under the Biological Diversity Act

Pre-Filing Confidence

Screens formulations against known legal bars before filing — reducing wasted attorney fees and doomed applications

Economic Impact

- Reduce IP research and legal costs
- Reduce avoidable patent rejections
- Faster and more confident IP/commercialisation decisions

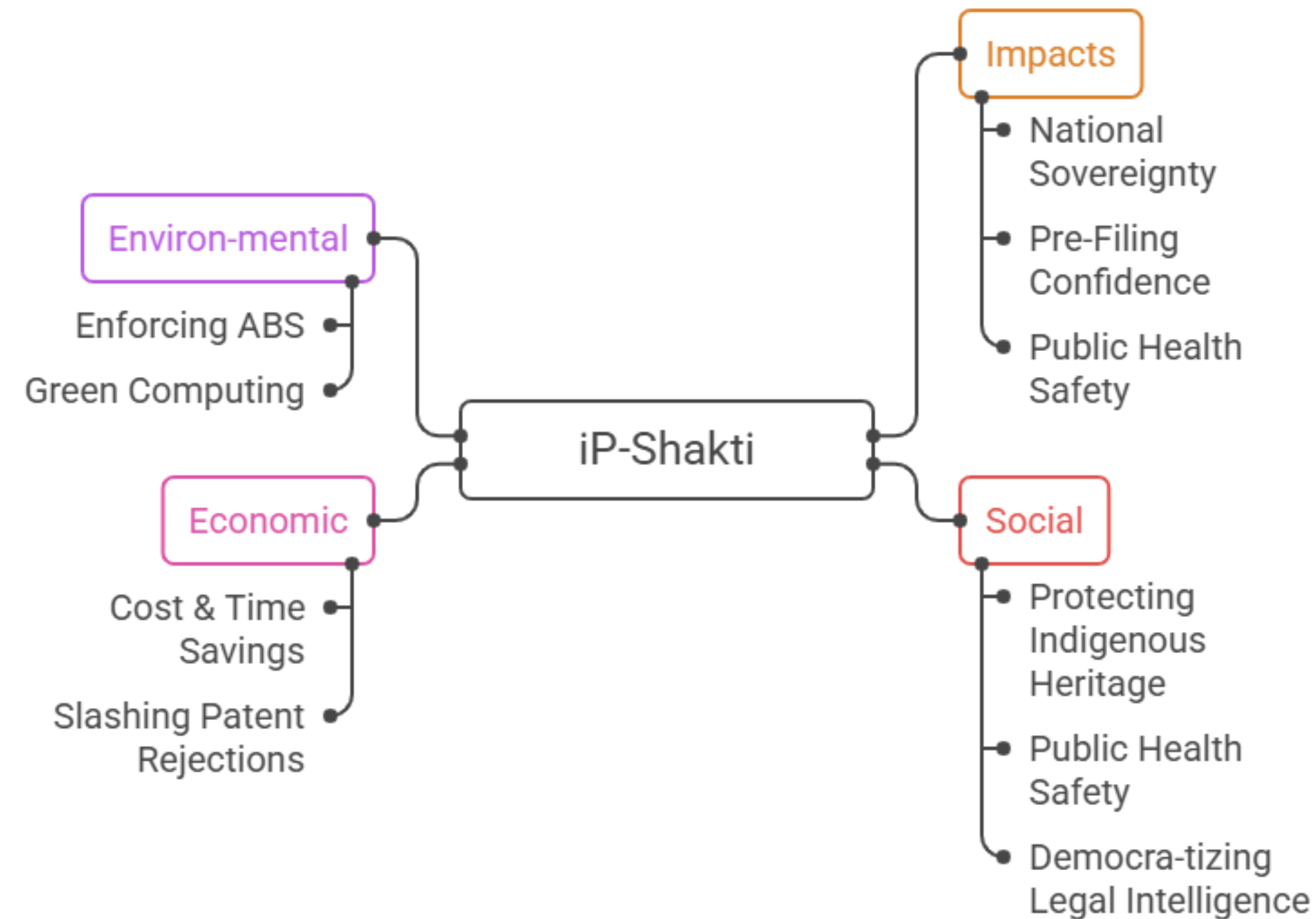
Social Impact

- Protect India's traditional knowledge
- Prevent biopiracy and misappropriation
- Empower Vaidyas, researchers and grassroots innovators

Environmental Impact

- Encourage responsible use of biological resources
- Support Access & Benefit Sharing (ABS)

@SIH15



A. Research

1. Corpus Research

- Traditional Knowledge Digital Library (TKDL) — tkdl.res.in
- Statutes & rules — India Code, indiacode.nic.in
- IP India public databases (patents/InPASS, trade marks, designs, GI Registry) — ipindia.gov.in
- National Biodiversity Authority / ABS — nbaindia.org

2. Technical Research

- [Anthropic Research \(Sept 2024\) — Contextual Retrieval: Improving RAG Accuracy and Minimizing Hallucination in Complex Corpora.](#)
- [Qdrant Hybrid Search & Cross-Encoder Architecture — Combining sparse BM25 and dense embeddings with rank-fusion](#)
- [Stanford LegalBench-RAG \(2024\) — Benchmark framework for snippet-level statutory retrieval and legal precision.](#)

B. References

- Legal: IP India • NBA • AYUSH • FSSAI • WIPO • WTO
- Traditional Knowledge: TKDL • CBD • Nagoya Protocol
- Technology: RAG Research • LangGraph • Qdrant